1 . What is airflow?

Apache Airflow is an open-source tool used to schedule, manage, and automate data workflows.

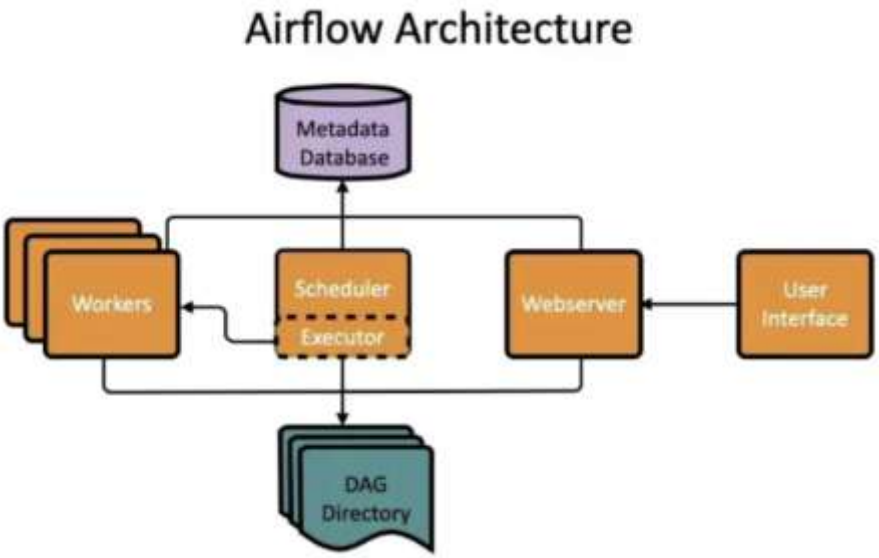
In simple words, Airflow helps you run multiple data related tasks automatically in a particular order without manually executing each task.

Working on a data engineering project we need to execute each task manually, one after another,

1. Extract data from MySQL.
2. Clean the data using Python.
3. Transform the data using PySpark.
4. Load the processed data into a database.
5. Generate a report.

But what if you want this process to run automatically every day at 9 AM?

That's where Airflow comes in!



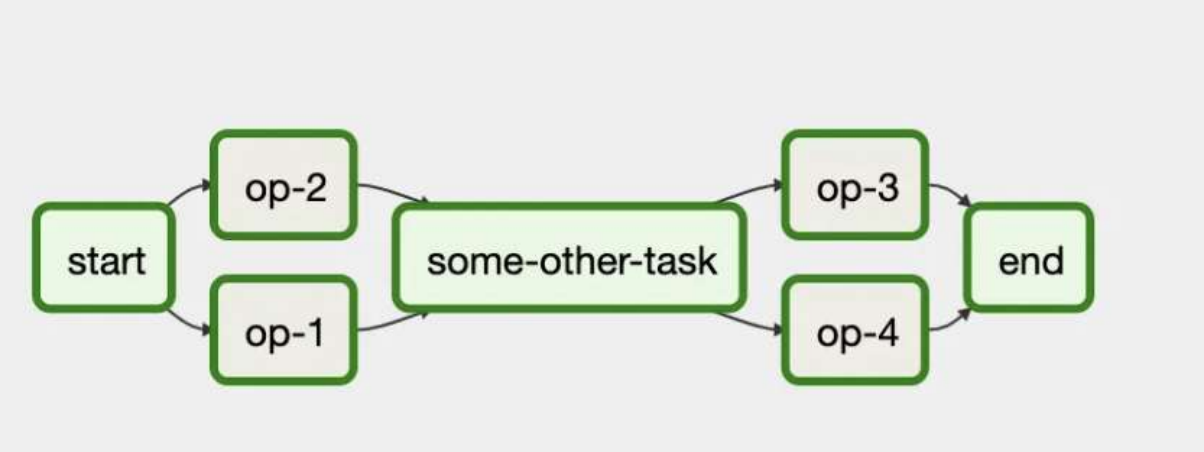
Airflow organizes workflows using something called a DAG (Directed Acyclic Graph).

2. DAG (Directed Acyclic Graph)

DAG, which stands for Directed Acyclic Graph represents the workflow you want to schedule and run. Each DAG is composed of tasks, which define the work to be done. The directed and acyclic nature of the graph means that the tasks flow in a specific direction (e.g., from one task to another), and there are no cycles (a task cannot depend on itself).

DAGs are defined in Python, and they are designed to be dynamic, meaning they can be programmatically generated, modified, or extended. This flexibility is one of Airflow's strongest features.

- Directed: Tasks follow a particular direction or order.
- Acyclic: Tasks cannot form a loop or cycle.
- Graph: Tasks are represented as nodes connected by relationships.



3. What is Task and Operator:

Task:

A Task is an individual unit of work or activity inside an Airflow DAG.

In simple words, a task represents one particular operation that needs to be performed.

Example: Consider a data pipeline:

1. Extract data from MySQL.
2. Clean the data.
3. Transform the data.
4. Load data into another database.

Each of these four operations is a separate task.

Operator:

An Operator defines what type of work a task should perform.

In simple words, an operator is a template or instruction that tells Airflow how to execute a task.

For example:

- PythonOperator – Executes a Python function.
- BashOperator – Executes a Bash command.
- SQLOperator – Executes SQL operations (specific SQL operators depend on the provider).
- EmailOperator – Sends emails.

4. Advantages of Apache Airflow

1. Workflow Automation
Automates repetitive tasks without requiring manual execution.
Example: Automatically extract data from MySQL every day.

2. Task Scheduling
Allows you to schedule workflows at specific times or intervals.
Example: Run a data pipeline every day at 9 AM.

3. Task Dependency Management
Executes tasks in the correct order based on dependencies.
Example: Data transformation starts only after data extraction is completed.

4. Monitoring and Tracking
Provides a web-based dashboard to monitor workflow execution, task status, and errors.

5. Error Handling and Retry Mechanism
If a task fails, Airflow can automatically retry it based on configured settings.

6. Scalability
Can manage simple workflows as well as complex data pipelines involving multiple tasks and systems.

7. Integration with Multiple Technologies
Works with Python, SQL, MySQL, PostgreSQL, Spark, cloud platforms, and many other tools.

8. Open Source and Free
Apache Airflow is open-source, so you can use it without purchasing a software licence.
Infrastructure costs may still apply.

9. Reusability
Workflows can be defined using Python code, making them easier to modify and reuse.

10. Improved Productivity
Reduces manual work and allows data engineers to focus on data processing and analysis.